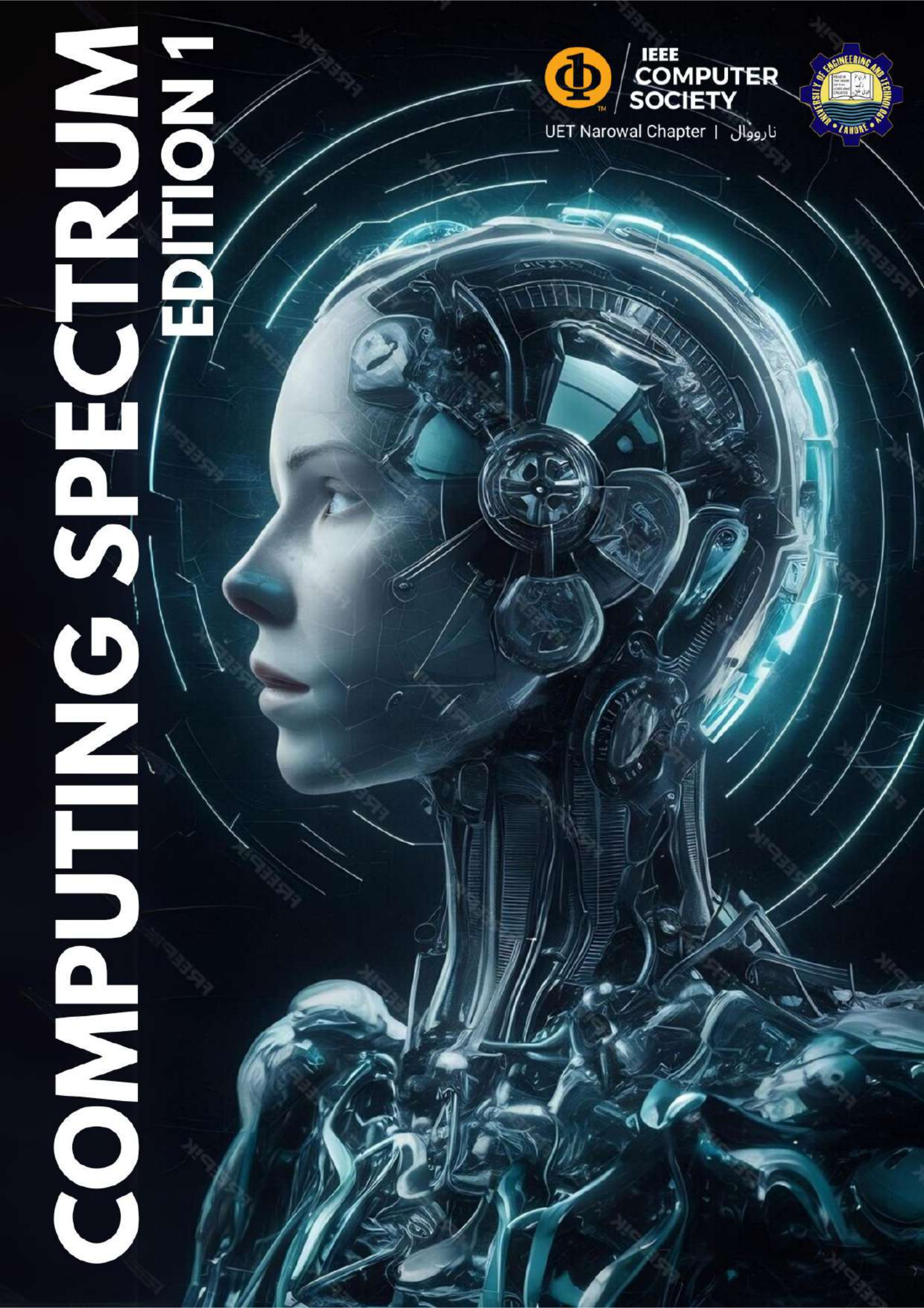

COMPUTING SPECTRUM

EDITION 1

IEEE COMPUTER SOCIETY

UET Narowal Chapter | نارووال

UNIVERSITY OF ENGINEERING AND TECHNOLOGY LAHORE

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Message from Campus Coordinator

Prof. Dr. Muhammad Shahbaz

It gives me immense pleasure to congratulate the entire team behind the first edition of the IEEE Magazine. This milestone marks not just a beginning, but a testament to the passion, creativity, and dedication of our students and faculty specially Ms. Fatima who have worked diligently to bring this initiative to life.

This magazine serves as a powerful platform to showcase the talent, innovation, and intellect of our students. I am positive that it will reflect the vibrant academic culture we strive to foster at our campus; one that values curiosity, encourages inquiry, and celebrates achievement. I firmly believe that such student-led initiatives play a critical role in nurturing leadership, collaboration, and communication skills that are essential for professional growth.

I have high hopes and expectations from the IEEE student body. You represent the future of technology, and your contributions will shape not only the future of our university but also the betterment of society at large. I urge you to continue thinking beyond boundaries, embracing challenges, and using your knowledge to make meaningful impacts.

Remember, excellence is a journey, not a destination. Strive to uphold the values of integrity, perseverance, and innovation in all that you do. Let this magazine be a beacon of inspiration for others and a symbol of what we can achieve together when we work with purpose and unity.

Once again, congratulations to the IEEE Magazine team on this remarkable achievement. May this be the first of many milestones to come.

Message from Advisor

Fatima Shahzadi

The IEEE Computer Society (IEEE-CS) chapter was officially established in September 2024, and I am honored to serve as the chapter's advisor. Over the past year, the society has achieved remarkable milestones by laying a strong foundation through the conduction of workshops, webinars, expert talks from industry leaders, and outreach programs. These initiatives have not only offered students exceptional opportunities to explore, learn, and innovate beyond the classroom but have also enabled our core team and management to delve into technical advancements alongside professional development.

I am sincerely grateful to all the professionals who generously contributed their time and expertise by conducting workshops and seminars as guest speakers. Their commitment to staying connected with our chapter is deeply appreciated.

It is with great pride that I present this inaugural edition of our publication, marking yet another significant step in the continued growth of our society. In future editions, we aim to highlight entrepreneurial journeys and innovative contributions from our recently graduated alumni and on-campus students who have significantly impacted the community.

I extend my heartfelt best wishes to the entire IEEE-CS chapter team and leadership. I am confident that through collaboration and dedication, you will continue to inspire and shape the next generation of innovators.

Message from Coordinator

Sadia Tariq

As the advisor of the IEEE Computer Society Chapter, I am immensely proud to witness the growth and enthusiasm of our talented student members. This chapter serves as a vibrant platform for innovation, collaboration, and technical excellence. Each event and initiative during the past year reflects their dedication and skill that defines future leaders in technology. I encourage all members to remain vigilant of technological advances and keep pushing boundaries. Together, we will shape a smarter future.

IEEE Computer Society

The IEEE Computer Society Narowal Chapter is a dynamic and rapidly growing student-led body under the umbrella of the IEEE UET Lahore Narowal Campus Student Branch.

Established with the mission to bridge the gap between academic learning and real-world technological advancements, the chapter serves as a vibrant platform for aspiring computer scientists, engineers, and tech enthusiasts.
Driven by innovation, collaboration, and excellence, the chapter organizes a variety of technical workshops, seminars, hackathons, coding competitions, and industrial collabs to empower students with practical skills and industry-relevant knowledge. It fosters an inclusive environment where students can explore diverse fields such as artificial intelligence, cybersecurity, software engineering, data science, and more.

As a part of the globally renowned IEEE Computer Society, the Narowal Chapter aims to uphold the highest standards of professionalism and academic integrity. It also encourages leadership, teamwork, and community engagement, equipping members with not just technical proficiency, but also the confidence to lead and make a difference.

With a strong commitment to growth and innovation, the IEEE Computer Society Narowal Chapter continues to inspire a new generation of thinkers, problem-solvers, and changemakers in the field of computing.

Department of Computer Science & Engineering

Department of Computer Science and Engineering, Narowal Campus started working in 2014 at UET Lahore, Narowal. Here the students are given the sterling opportunity to get education of highest standards in pleasant and friendly atmosphere, and to make compatible with their means. The students are laudably facilitated with Well-equipped Laboratories, state of the art Computer Labs with Internet Facility, Multimedia Assisted Classrooms, Innovation Center, unlimited Scholarships and consistently Updated Library. One of the most admirable features of Computer Science and Engineering Department at Narowal campus is its Highly Educated Faculty.

Computing Spectrum

Department Spotlights

At UET Narowal, our students consistently shine in both academic and co-curricular arenas, making their mark on national stages. In this edition, we proudly spotlight the outstanding achievements of students from Computer Science department who have brought honor to the university by securing top positions in prestigious competitions and representing UET Narowal at the national level.

Computing Spectrum

Computing Spectrum

Computing Spectrum is the official magazine of the IEEE Computer Society Narowal Chapter, created as a platform to showcase the creativity, knowledge, and technological insight of students, faculty, and professionals in the field of computer science and engineering.

Launched with the vision to inform, inspire, and innovate, this magazine serves as a reflection of the dynamic and ever-evolving world of computing. From thought-provoking articles on emerging technologies like artificial intelligence, machine learning, cybersecurity, and data science, to interviews, student research, event highlights, and creative tech content, Computing Spectrum offers a diverse range of perspectives and voices. This inaugural edition marks an important milestone for our chapter and our university, providing students with the opportunity to express their ideas, share their achievements, and connect with a wider community of tech enthusiasts.

Each page represents the dedication and collaborative spirit of our contributors, and we hope this magazine becomes a lasting source of inspiration and pride for current and future generations of innovators at UET Lahore Narowal Campus.

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ARTICLES & IDEAS

Low-Code/No-Code Platforms: Revolution or Risk?

Azhar Hayat

Low-Code and No-Code (LC/NC) platforms are steadily changing the way software is developed. These platforms make it possible for people with little or no coding experience to build applications through simple visual interfaces. Instead of writing long lines of code, users can drag and drop components, configure workflows, and connect data sources to create working software. This shift is seen by many as a step toward making technology development more
accessible and less time-consuming.

"The rise of low-code development platforms is changing the rules of who can build software"

At the same time, this growing trend raises some concerns. Developers and IT professionals often ask whether such platforms can really support complex or large-scale applications. Others worry about long-term maintainability, security, and vendor lock-in. In this article, we will look at both the potential advantages and the possible limitations of Low-Code and No-Code platforms. Low-Code and No-Code platforms have gained attention in recent years as businesses seek ways to digitize processes without hiring large software teams. Tools like Microsoft PowerApps, Google AppSheet, and OutSystems have become more common in workplaces, helping users automate tasks, build internal tools, and even create customer-facing applications.

One key reason for their rise is the global shortage of skilled software developers. By allowing non-developers to participate in app creation, these platforms can speed up delivery times and reduce development backlogs. Reports from research firms like Gartner predict that by 2025, a significant portion of all applications will be created using LC/NC platforms.

There are several reasons why organizations are turning to LC/NC tools:

- Faster Development: Applications can often be built and deployed in days instead of weeks.
- Lower Costs: With fewer developers required, the cost of building software can decrease.
- Broader Participation: Business users can contribute directly to development, which can improve alignment with business needs.
- Flexibility: Many platforms offer templates and integrations that make it easier to get started.

Some large organizations have successfully implemented LC/NC platforms for streamlining operations or integrating with existing systems, but they often do so with close collaboration between technical and non-technical teams. From a developer's point of view, LC/NC platforms are not necessarily a threat but can be a useful addition. They can help reduce repetitive work and free up time for more complex tasks. However, there is also the risk that apps built by non-developers will eventually need refactoring or re-engineering to meet performance and scalability needs. A balanced approach, where developers support or review LC/NC projects, might offer the best of both worlds. Low-Code and No-Code platforms offer a promising way to expand access to software development and speed up innovation. They can be very effective for simple tasks, quick solutions, and internal tools. Still, they are not without risks. Security, maintainability, and long-term scalability must be considered carefully.

In the end, these platforms are not a replacement for traditional development, but they can be a helpful complement when used thoughtfully. As the technology matures, organizations that take a balanced approach are more likely to benefit from the strengths of LC/NC tools while avoiding their potential downsides.

Are we Secure?

Fatima Shahzadi

The concept of security, traceable to 1900 B.C. with Egyptian scribes, has evolved far beyond a basic sense of protection. It has become a sophisticated system integral to cultural, military, and digital infrastructures. This progression from the Egyptian to the Mesopotamian civilizations, through the Roman era under Julius Caesar, represents a continuum of intellectual advancement and cultural exchange, culminating in the modern practices of cybersecurity.

Historically, cryptography emerged from military necessity, developed to secure sensitive communications. However, its applications soon extended beyond the battlefield. Cryptography not only safeguarded strategic information but also carried the intellectual imprints of civilizations, embodied in encrypted formulas, architectural blueprints, algorithms, and artifacts. These engravings on ancient slates chronicle the evolution of rational thought and cultural transformations across regions and eras.

The foundations of cryptography became a platform for intellectual discourse, enabling scholars and analysts to exchange ideas and ignite cultural revolutions. Egyptian scribes, esteemed within their society, used nonstandard hieroglyphs for inscribing messages on clay tablets—marking the earliest recorded use of cryptography. By 1500 B.C., Mesopotamian advancements had surpassed those of Egypt. A tablet discovered from this era contained an encrypted formula for pottery glazing, showcasing the use of symbolic variation in context-dependent meanings.

The Roman general Julius Caesar later introduced a monoalphabetic substitution cipher, shifting letters three positions to encrypt messages—a method now known as the Caesar Cipher. He further strengthened the cipher by substituting Greek characters for Latin ones, significantly enhancing the security of military and governmental communications.

Islamic scholars also made substantial contributions. In 725 A.D., Abu ‘Abdal-Rahman al-Khalil ibn Ahmad al-Farahidi authored a now-lost book on cryptography and famously solved a Greek cryptogram. In 855 A.D., Abu Wahshiyyaan-Nabati published various cipher alphabets used in encrypting magical formulas, demonstrating the broadening applications of cryptography beyond statecraft and into intellectual and mystical domains.

In 1790, Thomas Jefferson invented a wheel cipher composed of 26 rotating disks. Although underutilized during his lifetime, the device laid the groundwork for modern mechanical encryption tools. This concept was independently reinvented by Charles Babbage in 1854 and later refined in 1913.

The early 20th century witnessed a new era of cryptographic sophistication. During World War I (1914–1917), nations employed transposition and substitution ciphers for radio communications. The Allied cryptanalysis of the German Zimmerman Telegram, which revealed Germany’s proposal of a military alliance with Mexico, was pivotal in the United States' decision to enter the war.

Between 1939 and 1942, Allied forces famously broke the German Enigma cipher, a breakthrough that significantly shortened World War II. The U.S. National Security Agency later adopted the Lucifer cipher design as the Data Encryption Standard (DES) in 1976, gaining global acceptance.

In 1977, cryptographers Ronald Rivest, Adi Shamir, and Leonard Adleman introduced the RSA encryption algorithm—a groundbreaking public-key cryptosystem that enabled both confidentiality and digital signatures. RSA encryption remains foundational to today’s digital communication, bolstered by key lengths of 2048 and 4096 bits to ensure durability against brute-force attacks. Nevertheless, concerns persist about the longevity of this key extension approach, especially with the emergence of quantum computing.

While it is estimated that current supercomputers would require millions of years to breach RSA or AES with 2048-bit keys, quantum computers pose a significant threat. Quantum computing could potentially break these encryption standards within hours. In response, post-quantum cryptographic algorithms are under development to safeguard future communications. The commercial release of quantum computers has been gradual, with IBM officially introducing its quantum system at the end of 2024.

Echoing Edgar Allan Poe’s assertion in The Gold Bug—"Yet it may roundly be asserted that human ingenuity cannot concoct a cipher which human ingenuity cannot resolve"—the field of cryptography remains in constant evolution.

Despite the deployment of secure algorithms, data breaches continue to rise, affecting individuals, corporations, government entities, and healthcare systems. Since the final quarter of 2022, billions of bytes of sensitive information have been compromised. Thousands of malware variants inject malicious payloads daily, exploiting vulnerabilities in digital infrastructure.

Cyber warfare has become a defining element of modern conflict. In the Russia-Ukraine war, malware played a destructive role. In the Israel-Hamas conflict, digital intelligence was instrumental in high-profile attacks. During the Indo-Pakistan tensions, advanced sensor fusion technologies were used to disable aircraft—highlighting the extent of digital control in contemporary warfare.

Moreover, human activity is increasingly subject to surveillance. Digital transactions and online behaviors are constantly monitored, yet these data trails fail to capture emotional nuances such as love, anger, or deceit. As Shakespeare’s Macbeth aptly states, “Fair is foul, and foul is fair”—the digital realm often masks reality, where the innocent may be wrongly accused and the guilty may go unnoticed.

This dual nature of digital technology—both beneficial and threatening—defines our era. A fundamental question arises: Should individuals share passphrases with trusted persons or store them in browsers? Storing in browsers may compromise security, particularly when non-domestic servers are involved. Nations such as China and Russia have developed domestic browsers to ensure secure communication channels.

Yuval Noah Harari, in 21 Lessons for the 21st Century, warned of the power of data algorithms. In the chapter titled Big Data is Watching You, he references a future ”Yet the same Big Data algorithms might also empower a future Big Brother, so that we might end up with an Orwellian surveillance regime in which all individuals are monitored all the time [1]”. This warning underscores the pervasive sense of insecurity that underpins our increasingly digital world.

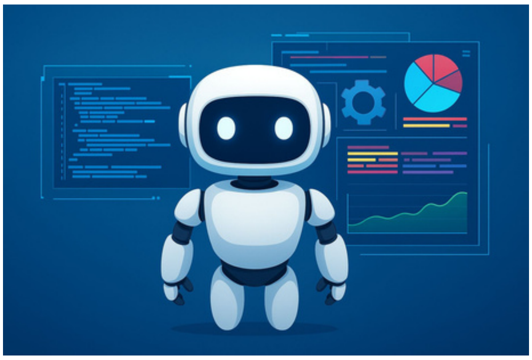
In conclusion, the journey of cryptography—from inscribed slates in ancient Egypt to post-quantum algorithms in the 21st century—reflects humanity’s enduring quest for secure communication. As we navigate this digital frontier, understanding both its potential and its perils remains essential to preserving freedom, privacy, and innovation.

Agentic AI: The Invisible Hand Behind Tomorrow's World

Muhammad Rafay Khokhar

Picture this: a world where decisions aren't just made, they're anticipated, strategized, and acted upon by intelligent systems that think for themselves. Welcome to the era of Agentic Artificial Intelligence. A powerful force quietly transforming our industries and reshaping our future.

"The development of full artificial intelligence could spell the end of the human race"



Unlike the AI of the past that simply followed orders, Agentic AI is different. It sees, thinks, and acts with purpose. It can assess its surroundings, make smart decisions, and take meaningful actions to achieve complex goals and it's all without constant human guidance. This shift from automation to autonomy is revolutionizing every corner of our lives. In healthcare, Agentic AI dives into massive datasets to predict illnesses before symptoms even appear, tailor treatments to individuals, and fast-track drug discovery. In finance, it keeps a sharp eye on transactions, sniffs out fraud in real time, and helps investors make better moves. City planners are using it to bring smart cities to life—with intelligent traffic systems, eco-friendly infrastructure, and faster emergency responses becoming the new normal. But with this exciting power comes responsibility.

If AI agents can make decisions on their own, who is responsible when things go wrong? How do we make sure their "thinking" stays aligned with human values and ethics? The truth is, Agentic AI isn't something we can afford to ignore. It's not just a shiny new tool, it's a game-changer. Those who embrace it thoughtfully will lead the next wave of innovation and prosperity. Those who don't? They'll be left scrambling to catch up. As we step into this new era, Agentic AI stands as both our partner and our challenge. It's shaping the future right now and it's up to us whether we ride the wave or watch it pass us by.Agentic AI marks a monumental shift from mere automation to true autonomy—AI that sees, thinks, and acts with intent. It is transforming industries, from diagnosing diseases before symptoms appear to building smarter, safer cities. But with its power comes an urgent need for responsibility, ethical alignment, and governance. As this intelligent force shapes our present and defines our future, we stand at a crossroads: embrace it thoughtfully and lead the next era of innovation, or hesitate and risk falling behind. Agentic AI is no longer science fiction—it's our reality, and it demands both vision and vigilance.

Light Cubes: Freezing Light into Crystals

Sadia Tariq

What if we could take a beam of light—intangible, weightless, and momentary—and trap it in a solid crystal? It sounds like science fiction, but researchers around the world are turning this amazing concept into scientific reality. The idea of “freezing light into crystal”; refers to the process of capturing and storing photons in solid-state materials, a technique that could not only revolutionize quantum computing and secure communication but also our understanding of light itself. As a wave, as a particle or as an amalgam of both!

At the heart of this phenomenon lies the field of quantum optics. Scientists use ultra cold crystals—often rare-earth materials cooled to near absolute zero—to create an environment where light can be slowed down, stopped, and stored. This isn’t just slowing light in the traditional sense, it is about embedding its quantum state within the atoms of a crystal. The photons essentially engrave their information onto the crystal, which can later be "read" and released as light again. It is a bit like turning a flash of light into a digital memory card. We may wonder why this is important to computers. Because light carries information incredibly fast, and being able to store and retrieve it opens the door to a relatively newer, quantum memory, which is a key component in building quantum networks. This memory is not about saving pictures or files; it is about storing quantum bits (qubits), which are the building blocks of the next generation of computers. The near ideal computers equipped with limitless processing speed and memory that can work non-deterministically!

While the process is still in the experimental stage due to technical hurdles such as short storage times and low efficiency, progress is quickening. Top notch Research Institutions like MIT, the University of Geneva, and others are exploring how different crystal structures can be optimized to hold light longer and more reliably.

In the future, we may witness crystals that not only sparkle but also hold the secrets of instant, like a time crystal. It will be able to promise secure communication and ultra-fast, never seen before kind of data processing. For now, freezing light into crystal remains one of the hottest and coolest topic in quantum physics.

ANCIENT TECHNOLOGY OF EGYPT

KHUBAIB AHMAD

Egypt, located in northeastern Africa, is one of the oldest and most influential civilizations in human history. It developed along the banks of the Nile River, which played a crucial role in its agriculture, economy, and culture.

Technological and Architectural Achievements

The Egyptians were highly skilled in engineering and mathematics, using these disciplines to create monumental structures. They developed a complex system of writing (hieroglyphs), and their advancements in medicine, such as surgeries and medical texts, were centuries ahead of their time.

The Pyramids of Giza

The Great Pyramid of Giza is the largest Egyptian pyramid. It served as the tomb of pharaoh Khufu, who ruled during the Fourth Dynasty of the Old Kingdom. Built c. 2600 BC,[3] over a period of about 26 years.

The pyramid is the oldest of the Seven Wonders of the Ancient World, and the only wonder that has remained largely intact. It is the most famous monument of the Giza pyramid complex, which is part of the UNESCO World Heritage Site "Memphis and its Necropolis".[5] It is situated at the northeastern end of the line of the three main pyramids at Giza.

Technology Behind the Pyramids

The Egyptians had mastered the use of copper tools, such as chisels and saws, to cut the limestone and granite blocks.

The alignment of the pyramids with celestial bodies (like the North Star) indicates their knowledge of astronomy and the use of simple tools like plumb bobs and leveling instruments for precision

Discovery of Hidden City

Italian and Scottish researchers claim they found a major discovery beneath the Pyramids of Giza, potentially rewriting the history of ancient Egypt. Using radar technology, the team led by Corrado Malanga from Italy's University of Pisa and Filippo Biondi from the University of Strathclyde in Scotland announced the findings of what they describe as a vast underground city stretching more than 6,500 feet directly beneath the pyramids."This groundbreaking study has redefined the boundaries of satellite data analysis and archaeological exploration," said the project's spokesperson, Nicole Ciccolo, according to The Sun.

She elaborated that the discovery "could redefine our understanding of the sacred topography of ancient Egypt, providing spatial coordinates for previously unknown and unexplored subterranean structures." The researchers used a new radar technology known as Synthetic Aperture Radar (SAR), which combines satellite radar data with tiny vibrations from naturally occurring seismic movements. The radar was used to create high-resolution three-dimensional images of underground structures, a method similar to using sonar to map the ocean floor. Their study, still awaiting peer review, suggests that the complex is ten times larger than the pyramids themselves.

The team claims to have discovered eight vertical cylinder-shaped structures, referred to as shafts, extending approximately 2,100 feet deep beneath the pyramids. Each shaft is said to be surrounded by spiral pathways that connect to two 80-meter cube-shaped structures. Above these, they reportedly found five multi-level structures connected by passageways.

"The existence of vast chambers beneath the earth's surface, comparable in size to the pyramids themselves, has a remarkably strong correlation with the legendary Halls of Amenti," Ciccolo stated. She explained that the cylindrical structures were found underneath each of the three pyramids and appear "to serve as access points to this underground system." Several experts have expressed skepticism regarding these claims. Independent experts, including Professor Lawrence Conyers, have raised serious concerns about the study. "I could not tell if the technology used actually picked up hidden structures below the pyramid," Conyers told Daily Mail, calling the claims of a vast city underneath the pyramids "a huge exaggeration."

He acknowledged that while small structures such as shafts and chambers might exist beneath the pyramids, the idea of a vast underground city is questionable. "The Mayans and other people in ancient Mesoamerica often built pyramids on top of the entrances of caves or caverns that had ceremonial meaning to them," he explained, according to The Sun.

"It is conceivable there are small structures, such as shafts and chambers, beneath the pyramids that existed before they were built because the site was special to ancient people."Critics point out that SAR technology typically excels at detecting shallower features, and its effectiveness diminishes significantly beyond a few meters in solid geological formations.

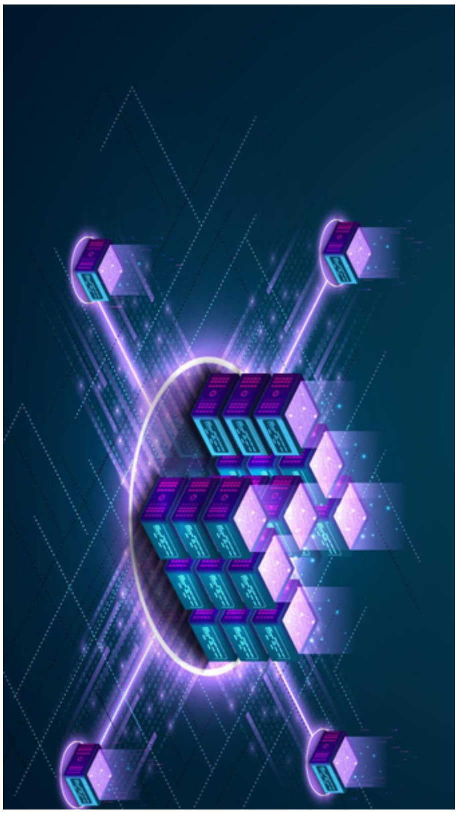
Despite the skepticism, the findings have sparked excitement online. Posts on social media platform X have fueled speculation, with some users suggesting that the structures could support alternative theories about the pyramids functioning as ancient energy systems rather than burial sites—a notion supported by figures such as Nikola Tesla and Christopher Dunn.

The legacy of ancient Egypt stands not only in its awe-inspiring monuments but also in the remarkable ingenuity of its people. From the precision of copper tools and the use of early surveying instruments, to their deep understanding of astronomy and architectural mastery, the Egyptians pushed the boundaries of what was possible with the knowledge and resources of their time.

The Pyramids of Giza remain a symbol of this brilliance—structures that continue to intrigue historians, engineers, and explorers alike. And as modern technology uncovers potential secrets beneath these ancient wonders, including the possible existence of a vast subterranean complex, the mysteries of Egypt remind us that the past still holds many untold stories. Whether fact or theory, each discovery fuels our curiosity and deepens our appreciation for one of humanity's greatest civilizations.

Blockchain: Securing Academic Credentials in Pakistan

Muhammad Abdullah



Universities in Pakistan award degrees and certificates to students when they graduate. These documents are proof of learning, but sometimes people lie and present fake certificates. The government has noticed such cases and taken action. For example, the Sindh Assembly told all universities to have every teacher's degree checked by the Higher Education Commission (HEC) to stop fake credentials(tribune.com.pk). In another case, a top official was fired when a re-check showed his claimed foreign degree was bogus (humenglish.com). These stories show we need a better way to protect and verify academic records.

What Is Blockchain?

Blockchain is a new way to record information that keeps it safe and unchangeable. Imagine a special notebook that everyone can see and use, where once you write something on a page, you seal the page so it can never be erased or changed. Blockchain works in a similar way, but it is digital. It is like a shared digital ledger (a record book) spread across many computers. Each "page" in this ledger is called a block, and each block is linked to the one before it, forming a chain. Once information is written in a block and added to the chain, it cannot be altered (scirp.org).

In simple terms, blockchain is like a notebook or a series of glass boxes where everyone can check the contents but nobody can sneak in and change past entries. Because copies of this notebook are held on many computers, if someone tries to cheat by changing their own copy, the other copies show the original record. This makes the system very secure and hard to fool.

Challenges with Academic Records in Pakistan

Pakistani universities and employers face several problems when verifying academic degrees:

• Fake Degrees and Scams: Some people use counterfeit diplomas or claim degrees from closed colleges. In one audit, a medical university in Sindh found a lecturer whose degree couldn't be confirmed; he was fired after the check (tribune.com.pk). This shows fake certificates can slip in.

• High-Profile Fraud: A well-known case involved the Director General of the national ID agency (NADRA). He claimed to have foreign degrees, but the foreign schools said he never attended them (humenglish.com). One of those schools had even shut down, so it had no records at all.

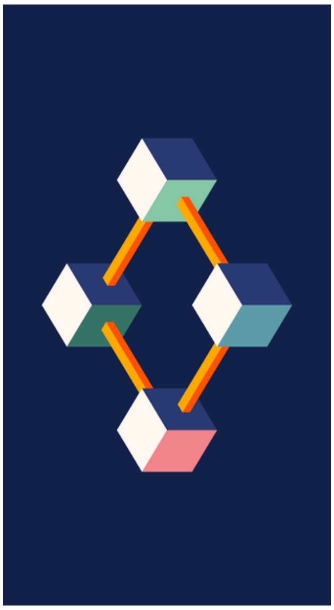
• Slow Verification: Checking a degree today means contacting the issuing university or using the HEC's online portal. This process can be slow and expensive, especially for foreign credentials. This delay can hurt students and employers.

• Cost and Trust: Verification often involves fees or long paperwork. Institutions have to trust a single source (like HEC's database), which can be problematic if that source is wrong or compromised.

How Blockchain Can Secure Academic Records

Blockchain offers a way to make academic records tamper-proof (meaning they can't be changed) and easy to verify. Here's how it helps:

- **Immutable Records:** Once a degree is added to the blockchain, it cannot be altered or deleted. It's like writing your diploma in a notebook and sealing that page forever. This means if a university writes your graduation record on blockchain, no one can later erase or fake it.
- **Shared Copies:** Many copies of the ledger exist across different computers (or "nodes"). Think of it like a Google spreadsheet that's shared with many people. When one copy gets updated, all copies update too. If someone tries to change their copy, it won't match the others, and everyone will see the mistake. This decentralization makes the record very trustworthy (scirp.org).
- **Instant Verification:** Employers or schools can verify a certificate quickly. For example, a graduate might get a digital diploma with a QR code or a special link. Scanning that code would bring up the record on the blockchain and show all the details immediately. No need to call a school or wait days.
- **Less Paperwork:** With blockchain, there is no need to send many requests and documents. A recruiter just checks the blockchain record directly. This can "reduce administrative work" and cut costs for universities and employers (scirp.org).
- **Preventing Fraud:** Because each record is linked and secured by cryptography (like a secret code), forging a certificate would break the chain. If someone creates a fake diploma, it won't have a matching entry on the blockchain, so it will fail the check. The chain itself "makes it difficult to manipulate the data" (scirp.org).



Example: Suppose a student finishes college. The college uses blockchain to record that student's degree.

The student gets a digital certificate with a QR code. Years later, when they apply for a job, the employer scans the code on their computer. Instantly, the employer sees on the blockchain that this student really did graduate from that college on that date. If someone else made a fake certificate for that college, there would be no matching blockchain entry, and the employer would know immediately it's fake.

Blockchain is already being explored for certificates in Pakistan. For instance, the Pakistan Blockchain Institute developed "Certificado," a system where universities issue diplomas on blockchain and students get a code to access their credentials (pakistanblockchaininstitute.org). This shows local efforts to use blockchain so diplomas can be verified instantly.

Comparing Verification Systems

Traditional Verification: Right now, verifying a certificate means relying on a central authority (like one computer at HEC or a university's office). If that central record is lost, hacked, or just slow, problems happen. It's like keeping all your money in one bank – if the bank has issues, you suffer.

Blockchain-Based Verification: With blockchain, there is no single point of failure. It's like having multiple vaults with the same lock. Records are decentralized copies are everywhere. Checking a degree becomes as easy as looking at a public list: you just scan the digital record and see instantly if it's valid. There is no middleman and no waiting.

Key differences include:

- **Where the Record Lives:** Traditional records live in one database; blockchain records live on many computers.
- **Speed:** Traditional checks (contacting HEC, universities) can take days; blockchain checks can be instant.
- **Trust:** Traditional requires trusting an institution or officials; blockchain relies on math and network consensus.
- **Security:** Traditional records can be forged or lost; blockchain records are locked together and very hard to fake.

In summary, Pakistan's universities and employers face a growing problem of fake or unverified degrees. Current verification methods are slow and sometimes unreliable. Blockchain offers a new solution: a tamper-proof way to store and check academic records. By using blockchain, diplomas become instantly verifiable and hard to forge.

Pakistan can benefit from this technology. If universities and HEC start using blockchain for certificates, students could carry digital diplomas that employers can trust with one scan. This would save time, reduce fraud, and make sure honest students' achievements are always recognized.

Convolutional Neural Networks: Shaping the Future of Deep Learning

Muhammad Zaid

In the vast landscape of artificial intelligence, Convolutional Neural Networks (CNNs) stand tall as one of the most influential innovations. Inspired by the human visual cortex, CNNs empower machines to see, recognize, and understand images in ways previously thought impossible. From facial recognition on smartphones to diagnosing diseases through medical imaging, CNNs have become the backbone of modern computer vision. As we step into 2025, CNNs continue to evolve, facing new challenges and adapting to the rise of architectures like Vision Transformers (ViTs). For beginners embarking on their AI journey, understanding CNNs is both essential and exciting.

How CNNs Work: The Art of Seeing

A CNN processes an image much like we do—first noticing edges, then shapes, and
eventually objects. It achieves this through layers:

- Convolutional Layers apply filters to detect simple patterns (like edges).
- Pooling Layers reduce the size of the data, focusing on the most important features.
- Fully Connected Layers combine everything to make the final prediction.

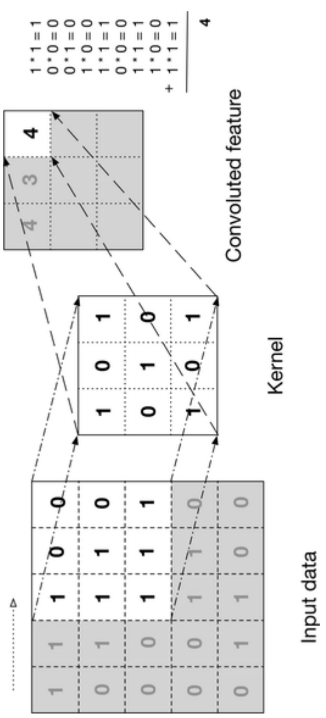
Each layer learns to identify more complex features as the data passes through. Imagine teaching a child to recognize a cat: first noticing whiskers, then ears, then combining these observations into a full understanding. CNNs learn similarly, but through mathematical optimization.

Diagram Suggestion: A simple visual of an input image flowing through convolution, pooling, and dense layers to final classification (e.g., cat vs dog).

Latest Trends: The New Age of CNNs

While CNNs have achieved incredible feats, innovation hasn't stopped. Recent trends
reshaping CNNs include:

- **Efficiency and Lightweight Models:** CNNs are becoming more compact and faster. Architectures like EfficientNetV2 and MobileNetV4 allow CNNs to run on smartphones, drones, and even wearables, without sacrificing much accuracy.
- **Hybrid Models (CNN + Transformers):** The rising success of Vision Transformers (ViTs), which use attention mechanisms instead of convolutions, has challenged the dominance of CNNs. However, CNNs are adapting. New hybrid architectures, like CoAtNet and ConvNeXtV2, blend the strengths of convolutions (local feature extraction) with transformers (global context modeling).
- **Self-Supervised Learning**: CNNs are learning without requiring millions of labeled images. Techniques like SimCLR and DINOv2 allow CNNs to pre-train on unlabeled data, making them more accessible and efficient.
- **Explainability and Trust:** Visualization techniques such as Grad-CAM and saliency maps help us understand what parts of an image the network focuses on.



CNNs vs. Vision Transformers: Friends or Rivals?

A burning question in 2025: Are CNNs becoming obsolete compared to Vision Transformers?
Short answer: No.

While ViTs dominate tasks with massive datasets, CNNs remain unbeatable when data is scarce or when fast, efficient inference is needed. CNNs are inherently better at recognizing local patterns with fewer examples.
Today's best models often combine both. For example, Google's CoAtNet starts with convolution layers to capture fine details, then switches to attention layers.
Diagram Suggestion: A side-by-side comparison showing CNNs (local pattern focus) vs Transformers (global attention focus).

Real-World Applications: Where CNNs Rule

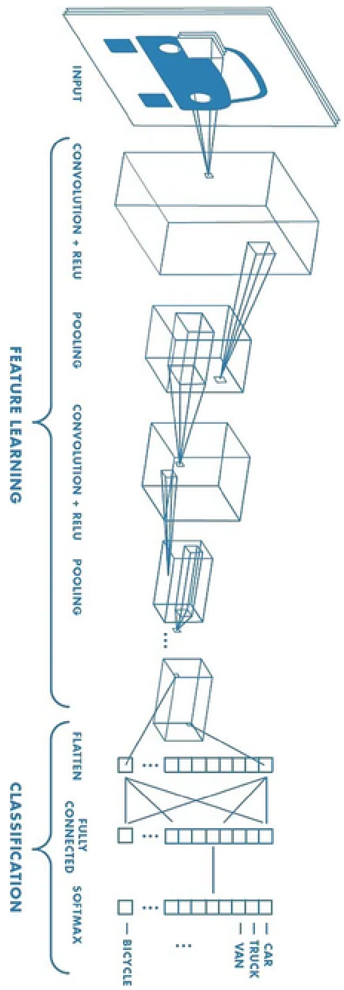
CNNs are embedded everywhere:

- Medical Imaging: CNNs detect tumors, fractures, and retinal diseases with near-human accuracy.
- Autonomous Vehicles: Self-driving cars use CNNs to recognize lanes, pedestrians, and obstacles.
- Agriculture: CNNs monitor crop health from satellite imagery.
- Security: Face recognition systems in airports and banking.
- Augmented Reality (AR): Real-time scene understanding in AR glasses.

The Future Outlook: What Lies Ahead

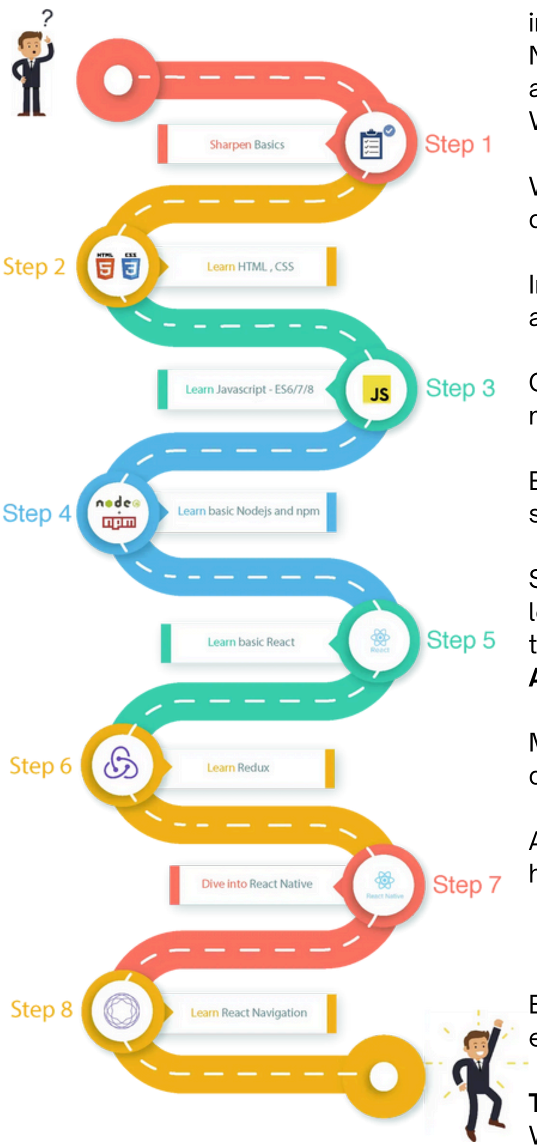
- The journey of CNNs is far from over. Here's what the future holds:
- Adaptive CNNs: Dynamic models that adjust their complexity based on input difficulty.
- Green AI: Energy-efficient CNNs to minimize carbon footprint.
- Multi-modal Learning: CNNs will collaborate with text, audio, and sensor data.
- Democratization: Easier tools for non-experts to build real-world CNN applications.

Despite the excitement around Transformers, CNNs' intuitive structure, lower data demands, and continual evolution ensure they will remain central to deep learning's future.



React Native

Abdul Ahad



React Native is a JavaScript framework that combines the best parts of native development with React, an important library for building user interfaces. Unlike traditional hybrid apps based on Web Views, React Native renders components using native APIs, delivering performance and user experience comparable to native applications.

Why React Native?

- **Cross-Platform Development:**

Write once, and deploy on both iOS and Android, reducing development time and resources.

- **Hot Reloading:**

Instantly view changes in the code without recompiling the entire application, speeding up the development.

- **Component-Based Architecture:**

Create encapsulated components that manage their state, resulting in more maintainable and scalable codebases.

- **Access to Native Modules:**

Easily integrate native code when needed, enabling the use of platform-specific features.

- **Declarative Syntax:**

Simplify the development process by describing what the UI should look like for a given state, making the code more predictable and easier to debug.

Advantages Over Traditional Development

- Cost Efficiency:

Maintain a single codebase for multiple platforms, reducing development and maintenance costs.

- Faster Time-to-Market:

Accelerate the development process with reusable components and hot reloading.

- Consistent User Experience:
- Ensure uniformity across platforms, enhancing user satisfaction.
- Vibrant Community:

Benefit from a large, active community that contributes to a rich ecosystem of libraries and tools.

The Future of React Native

With continuous latest updates and a strong backing from Meta, React Native is ready to remain a dominant force in mobile development. The introduction of the new architecture brings full support for modern React features, including Suspense, Transitions, and automatic batching, further enhancing performance and developer experience.

How Bitcoin works

Talha Amir

As the world continues to embrace Technology and the Internet more and more people and by extension businesses are doing transactions digitally. Now Imagine you want to do a transaction but you don't want the money to pass through some bank or middlemen and you do not trust any centralized system. You simply want the money to arrive, safely and securely. But without a central authority like a bank keeping records, how do you make sure no one cheats? How do you ensure the same money isn’t spent twice?
This is the fundamental problem Bitcoin set out to solve. In 2008, an unknown figure (or group) called Satoshi Nakamoto published a whitepaper proposing a system where money could be sent peer-to-peer without relying on any centralized institution. This system became Bitcoin — the first decentralized digital currency.

The Big Idea Behind Bitcoin
At its heart, Bitcoin answers a simple question:
Can we build trust without trusting any single person?

- Instead of relying on banks to record transactions, Bitcoin uses a distributed network of computers (called nodes) that all maintain a shared public ledger — known as the blockchain.
- Every transaction ever made is recorded on this ledger.
- Everyone can see it.
- No one can secretly alter it without the network noticing.

How It Works: Building Blocks of Bitcoin
Let’s break Bitcoin down into a few key ideas:
Blockchain: The Record Book

- Think of the blockchain like a linked list of blocks.
- Each block contains a bundle of recent transactions.
- Every new block points back to the previous one, forming a secure, chronological chain.

Because everyone keeps a copy of this chain, altering past records would require changing every copy across the entire network — nearly impossible.

Proof of Work: Solving Puzzles to Earn Trust
When new transactions happen, they aren’t added to the blockchain immediately. Instead, miners — special nodes on the network — race to solve a computational puzzle based on the new block's data.
The puzzle is hard to solve but easy to verify once completed.
The first miner to solve it gets to add the new block to the blockchain — and earns a reward in newly created Bitcoin.
This "puzzle-solving" is called Proof of Work and is what keeps the network secure and tamper-resistant.

Hashing: The Digital Fingerprint

A hash is a fixed-size string generated from data, kind of like a unique fingerprint. Bitcoin uses a cryptographic hash function (SHA-256) to:

- Link blocks together.
- Verify transaction integrity.
- Ensure tampering is easily detectable.
- Change even a single character in a transaction, and its hash completely changes.

“We're trying to create a comprehensive Bitcoin and crypto ecosystem so that we can grab the mantle of the most crypto-friendly city in the U.S.”

Francis X. Suarez

Peer-to-Peer Network: No Bosses, Just Peers

There’s no central server in Bitcoin. Instead, thousands of nodes communicate and synchronize with each other directly.

When a transaction is made:

- It’s broadcasted to the network.
- Miners pick it up, include it in a block, and secure it with Proof of Work.
- Nodes validate the block and add it to their copy of the blockchain.
- No one person is "in charge" — the system works through distributed consensus.

The Life of a Bitcoin Transaction

Let’s walk through a simple journey:

- You create a transaction sending Bitcoin to a friend.
- Your transaction is broadcast to the network.
- Miners pick up your transaction and many others, gathering them into a block.
- They compete to solve a Proof of Work puzzle.
- The first miner to solve the puzzle broadcasts their block.
- Other nodes verify it.
- Your transaction becomes a permanent part of the blockchain!

I**s Bitcoin a Viable Currency?**

Bitcoin has revolutionized how we think about money and trust. It proves that decentralized money is possible without banks or governments overseeing every transaction.

But Bitcoin also faces real challenges:

- Scalability: Bitcoin processes about 7 transactions per second. Compare that to Visa’s 24,000 per second.
- HFT: High frequency trading is also not possible on the blockchain as each block is added after roughly 10 minutes and typically after 6 or so blocks we trust a blockchain so HFT is not possible.
- Energy Consumption: Proof of Work consumes massive amounts of electricity.
- Volatility: Bitcoin's price swings wildly, making it difficult to use for everyday purchases.
- Regulatory Uncertainty: Governments are still figuring out how (or whether) to regulate cryptocurrencies.

In my view, Bitcoin is a breakthrough innovation, but it may not replace everyday currencies in its current form. Instead, Bitcoin feels more like digital gold — a store of value — rather than a day-to-day payment method.

Future technologies like the Lightning Network (for faster Bitcoin payments) or newer cryptocurrencies could bridge this gap. Still, Bitcoin laid the foundation for an entire new financial world — and that alone secures its place in history.

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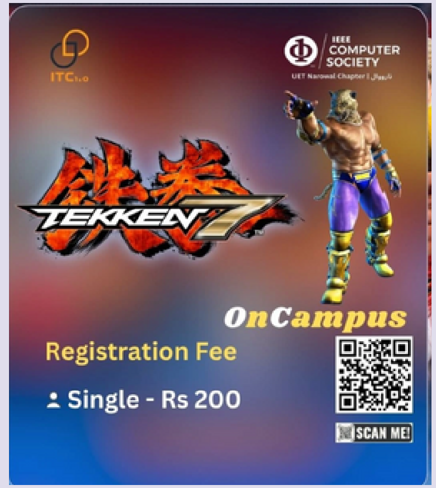

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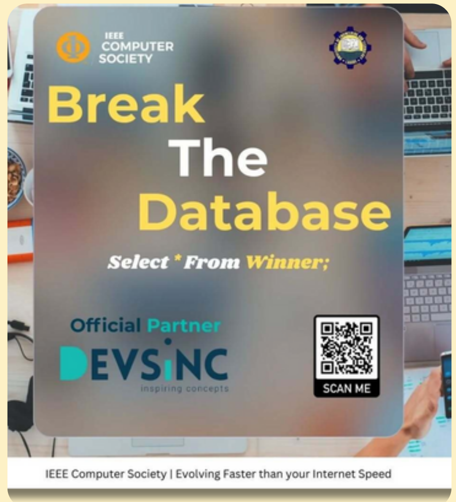

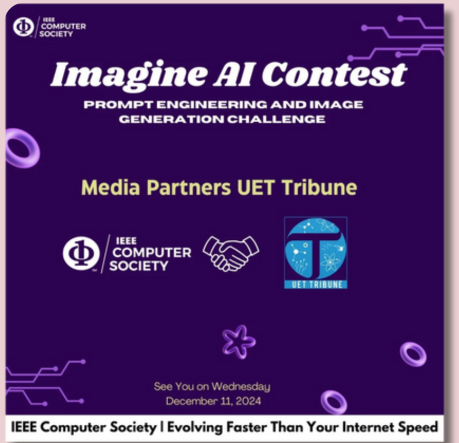

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